

DATA CLEANING & REPORTING AUTOMATION PROJECT REPORT

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Introduction

Data cleaning and reporting automation focuses on improving data quality and reducing manual effort in reporting workflows. The project uses Power BI and Excel to automate data preprocessing and dashboard generation.

Objectives

Handle missing values, remove duplicates, standardize inconsistent data, automate reporting, and improve reporting efficiency.

Tools Used

Microsoft Excel, Power BI Desktop, Power Query, Data Visualization.

Dataset Description

The dataset contains employee information including Employee ID, Name, Department, Salary, and City. It includes missing values, duplicate records, and inconsistent entries for cleaning practice.

Methodology

Import data, clean missing values, remove duplicates, standardize text fields, create KPIs, build dashboards, and automate refresh workflows.

Data Cleaning Process

Missing values were handled using replacement techniques. Duplicate employee IDs were removed. City names were standardized and extra spaces were trimmed. Data types were validated before loading.

Dashboard Visualizations

KPI Cards, Department-wise Bar Chart, City-wise Pie Chart, Salary Analysis Charts, and Interactive Slicers.

Automation

Power Query stores transformation steps. When new records are added and the dataset is refreshed, all cleaning operations and reports update automatically.

Business Insights

Employee distribution by department and city was analyzed. Data quality improved after cleaning, leading to more accurate reporting.

Conclusion

The project successfully automated data cleaning and reporting tasks, improving efficiency and reducing manual effort.